

Sheet G06 - Differential Geometry

Solutions for the theoretical and practical part via eCampus by Mo, 24.11.2025, 10:00.

Practical Part

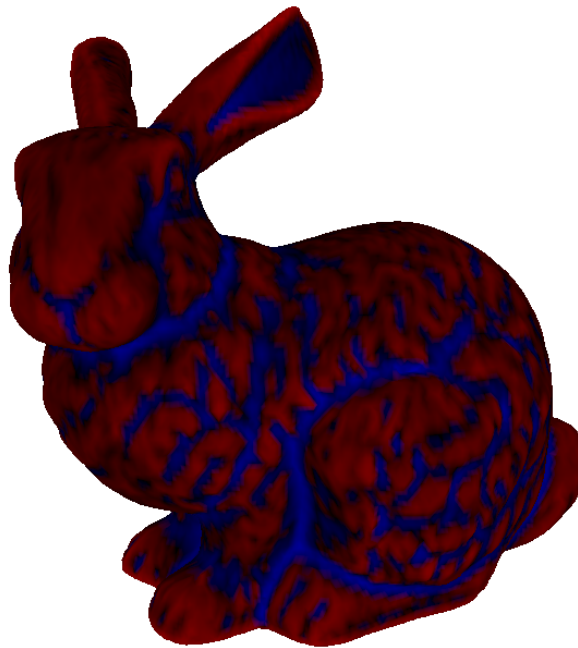


Figure 1: Mean curvature of bunny mesh.

Assignment 1) Curvature on Meshes

(10Pts)

Disclaimer: This exercise may be harder than the previous ones. So don't start too late and ask questions if you have major problems with the task.

In the lecture and this exercise, you have seen how the fundamental forms can be used to calculate properties of a parameterized surface. In this exercise, we will use this knowledge to calculate the curvature of a mesh at each vertex. The problem here, however, is the fact that a mesh is not a parameterized surface but a structure of discrete planar triangles. Therefore, we will use an approximate solution based on the algorithm of Rusinkiewicz [1].

Download the zip archive `framework_g06.zip` from eCampus and include it into the project.

- a) Read and understand the paper. You have to implement the algorithm at the beginning of section 3.4.

- b) Calculate all the properties needed for the linear system of equations that is described on top of page 3. As orthonormal coordinate frame (u, v) , you can use the triangle's edges and Gram-Schmidt. Then construct and solve the linear system of equations.
- c) Average the face tensors to the adjacent vertices. Rotate the face's coordinate frame onto the frame of the current vertex and project the tensor via Equation (7) in the paper.
- d) Compute the eigenvalues of the second fundamental form to get the principal curvatures.
Hint: For 2×2 matrices, there exists a closed form solution for eigenvalues.

Hint: If the algorithm takes too long, run the code in Release build.

Your final solution should look like Figure 1.

Submit this exercise by uploading the whole source file onto eCampus!

Theoretical Part

Upload the solution as pdf file on eCampus!

Assignment 2) Fundamental Form

(6Pts)

Let $u, v \in \mathbb{R}$. Consider the surface

$$q(u, v) = \begin{pmatrix} u \\ v \\ u^2 + 2v^2 \end{pmatrix}$$

- a) Calculate the first fundamental form of the surface $\mathbf{I}(u, v)$.
- b) Use $\mathbf{I}$ to calculate the area of the surface in the area $[0, 1]^2$.
Hint: Simplify the integral as much as possible. Then, you can use a calculator to get the numerical value.
- c) Calculate the second fundamental form $\mathbf{II}$ at the location $u = 1, v = 1$.
- d) Use $\mathbf{I}$ and $\mathbf{II}$ to calculate the Gaussian curvature. What does this value tell you about the properties of the surface at this point?
- e) Use $\mathbf{I}$ and $\mathbf{II}$ to calculate the principle curvature and principal directions. How can you compute the mean curvature without evaluating the integral? Calculate the numerical value.

$$\kappa_M = \frac{1}{\pi} \int_0^\pi \kappa_n(\mathbf{w}(\theta)) d\theta$$

References

- [1] Szymon Rusinkiewicz. "Estimating curvatures and their derivatives on triangle meshes". In: *Proceedings. 2nd International Symposium on 3D Data Processing, Visualization and Transmission, 2004. 3DPVT 2004*. IEEE. 2004, pp. 486–493.

Good luck!